

Course-End Project

Employee Performance Mapping

Overview

In this project, you will apply the fundamentals of SQL programming to practice, from data preparation to subqueries and functions. The focus is on building an employee performance mapping solution for a business startup. You will use SQL to map employees and track their performance accurately.

Instructions

- Review the learning materials in the SQL course
- Carefully read the situation, tasks, actions, and result sections to grasp the assignment fully
- Complete and submit your assignment via the Learning Management System (LMS)
- Follow the guidelines provided closely, ensuring your report includes all required analyses and interpretations

Scenario

ScienceQtech is a startup specializing in data science. The company has worked on projects involving fraud detection, market basket analysis, self-driving cars, supply chain optimization, algorithmic early detection of lung cancer, customer sentiment analysis, and drug discovery. With the annual appraisal cycle approaching, the HR analysis department has tasked you, a junior business analyst, with generating reports on employee details, performance, and projects undertaken. You are required to analyze the employee database and extract specific information based on various business requirements.

Tasks

To support the HR analysis department in finalizing employee performance mapping:

- Use manager-provided performance ratings to assess employee contributions
- Identify the maximum salary among employees and ensure that all job roles align with the organization's standard job profiles
- Calculate bonuses to estimate additional costs and determine which employees require training to enhance their performance

This analysis will support informed decision-making and improve overall organizational effectiveness.

Input Dataset

emp_record_table: It contains information on all employees.

- EMP_ID – ID of the employee
- FIRST_NAME – First name of the employee
- LAST_NAME – Last name of the employee
- GENDER – Gender of the employee
- ROLE – Post of the employee
- DEPT – Field of the employee
- EXP – The employee's years of experience
- COUNTRY – The employee's current country of residence
- CONTINENT – The employee's continent of residence
- SALARY – Salary of the employee
- EMP_RATING – Performance rating of the employee
- MANAGER_ID – The manager mapped to the employee
- PROJ_ID – The project on which the employee is working or has worked on

Proj_table: It contains information about the projects.

- PROJECT_ID – ID for the project
- PROJ_Name – Name of the project
- DOMAIN – Field of the project
- START_DATE – Day the project began
- CLOSURE_DATE – Day the project was or will be completed
- DEV_QTR – Quarter in which the project was scheduled
- STATUS – Status of the project currently

Data_science_team: It contains information about all employees in the Data Science team.

- EMP_ID – ID of the employee
- FIRST_NAME – First name of the employee
- LAST_NAME – Last name of the employee
- GENDER – Gender of the employee
- ROLE – Post of the employee

- DEPT – Field of the employee
- EXP – Years of experience the employee has
- COUNTRY – Country in which the employee is presently living
- CONTINENT – The continent in which the country is

Actions

1. Create a database named *employee*, then import **data_science_team.csv**, **proj_table.csv**, and **emp_record_table.csv** into the **employee** database from the given resources
2. Write a query to fetch **EMP_ID**, **FIRST_NAME**, **LAST_NAME**, **GENDER**, and **DEPARTMENT** from the employee record table, and make a list of employees and details of their department
3. Write a query to fetch **EMP_ID**, **FIRST_NAME**, **LAST_NAME**, **GENDER**, **DEPARTMENT**, and **EMP_RATING** if the **EMP_RATING** is:
 - less than two
 - between two and four
 - greater than four
4. Write a SQL query to retrieve the **employee ID**, **first name**, **role**, and **department of employees** who hold leadership positions (Manager, President, or CEO)
5. Write a query to list employee details, including **EMP_ID**, **FIRST_NAME**, **LAST_NAME**, **ROLE**, **DEPARTMENT**, and **EMP_RATING**, grouped by department. Also include the respective employee rating along with the max emp rating for the department
6. Write a query to find the minimum and maximum salary for employees grouped by their role, using data from the employee record table
7. Write a query to assign a rank to each employee based on their years of experience, using data from the employee record table
8. Write a query to create a view that displays employees in various countries whose salary is more than six thousand, using data from the employee record table
9. Write a query to create a view that lists employees across different countries who earn a salary greater than 6000, using data from the employee record table

10. Create an index to improve the cost and performance of the query to find the employee whose **FIRST_NAME** is **Eric** in the employee table, after checking the execution plan
11. Write a query to calculate the average salary distribution based on the continent and country. Take data from the employee record table

Result

Save your SQL queries with screenshots of the output in a Word document. Upload the solution document in the Learning Management System (LMS).